ENGINEERING

Engineering Onboarding Guide

Architecture, repositories, local setup, how we ship, on-call and your first-week quests.

Version	Owner	Effective from	Next review	Applies to
5.4	Jordan Lee, Engineering Manager	15 Sep 2026	15 Dec 2026	Engineers and ML engineers

1. Architecture at a glance

Service	Language	What it does
gateway	Rust	OpenAI-compatible endpoint. Routes customer model calls, applies the active model version per task, records traces.
calibrator	Python	Attributes outcomes to the experience used, and proposes memory, prompt and skill candidates.
forge	Python	Trains task-specific adapters on open-weight base models, including Banyan.
evals	Python	Runs candidates on held-out tasks, gates activation, and handles rollback.
console	TypeScript (Next.js)	The customer-facing app at app.mubit.ai.
infra	Terraform	Cloud infrastructure, networking and secrets wiring.

Traces are stored in Postgres with large payloads in object storage. Customer traces are Restricted data: see the Information Security Policy.

2. Local setup

```
git clone git@github.com:mubit-demo/monorepo.git
cd monorepo
make bootstrap   # installs toolchains with mise
make dev         # starts Postgres, Redis and MinIO in Docker
make seed        # loads synthetic fixtures, never customer data
make test        # unit and integration tests
```

If make bootstrap fails, ask in #eng-help before trying to fix the toolchain yourself.

3. How we ship

Branches and pull requests

- We use trunk-based development with short-lived branches named initials/ticket-slug, for example jl/RUN-412-retry-timeouts.
- Every pull request needs 1 approval. Changes to gateway, authentication or billing need 2.
- We squash-merge. Keep pull requests under about 400 changed lines where possible.

CI checks (all must pass)

- Lint and type checks.
- Unit and integration tests.

- Eval smoke test: a fixed 50-task held-out set. A change that drops task success by more than 1 point is blocked.
- Secret scanning with gitleaks, which also runs as a pre-commit hook.

Deploys

- Merging to main deploys automatically to staging.
- Production releases go out on the release train, Tuesday and Thursday at 14:00 UK time.
- Hotfixes outside the train need approval from the on-call lead.
- Launch week freeze, 26 to 30 October 2026: only launch-related changes, approved by the Head of Engineering.

4. On-call

You join the on-call rotation after 6 weeks, and shadow one full week first. Pages come through PagerDuty.

Severity	Definition	Acknowledge within
Sev1	Customer-facing outage, or any possible exposure of customer data	15 minutes
Sev2	Degraded service: higher latency or errors for some customers	1 hour
Sev3	Minor issue with a workaround	Next working day

Sev1 and Sev2 incidents get a blameless postmortem within 5 working days.

5. Secrets

Never put secrets in code, config files or Slack. Use 1Password for personal secrets and the secrets manager for services. If a secret leaks, rotate it immediately and report it in #security-incidents.

6. Your first-week quests

Day	Quest	Done when
1	Laptop, GitHub, Linear, Slack and 1Password set up	IT Helpdesk ticket closed
2	Run the local stack and the full test suite	make test passes on your machine
3	Ship your first pull request (a docs fix or a test)	Merged to main
4	Shadow a release train	You can explain each step of a release
5	Demo something at show-and-tell	Friday 16:00, five minutes